

7. Write a program that prints the addresses of code, global, static, heap, and stack variables. **Analyze** the Linux process address space layout and explain each memory segment.

Make use of the `/proc/<PID>/maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
advaith@DESKTOP-0E1OBMR: X + v
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int global = 100;
static int s = 50;

int main()
{
    int stack = 10;
    int *heap = (int*)malloc(sizeof(int));

    printf("Code Segment : %p\n", main);
    printf("Global Segment : %p\n", &global);
    printf("Static Segment : %p\n", &s);
    printf("Heap Segment : %p\n", heap);
    printf("Stack Segment : %p\n", &stack);

    free(heap);
    return 0;
}
```

Code:

```
advaith@DESKTOP-0E1OBMR:~$ nano heap.c
advaith@DESKTOP-0E1OBMR:~$ gcc heap.c -o program
advaith@DESKTOP-0E1OBMR:~$ ./program
Code Segment : 0x642d203f31a9
Global Segment : 0x642d203f6010
Static Segment : 0x642d203f6014
Heap Segment : 0x642d52e48010
Stack Segment : 0x7ffddf5ad87c
```

Output: